

An aerial photograph of Zurich, Switzerland, showing the city's dense urban landscape, the Limmat river, and various buildings. A large blue rectangular box is overlaid on the image, containing the title and project information.

System-level optimization for the integration of heat pumps in district heating and cooling

Final presentation for the MSc Thesis of **Konstantinos Furlaris**

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- 2) Project Supervisors: Dr. Tom Mike Terlouw, ETH Zurich, ESA Lab and PSI
Dr. Tarek Alskaif, Wageningen University & Research

- **Introduction**

- Methodology
- Results
- Outlook

Motivation



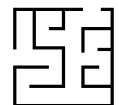
Observation:

- The residential heating and cooling sector contributes 35% of all energy related GHG emissions ^[1]
- Decarbonization of heating and cooling is on much slower pace than the electricity sector ^[2]



Status Quo:

- Heat Pumps, hot water energy storage and district heating and cooling (DHC) networks have emerged as decarbonization technologies for the heating and cooling sector ^[3,4]
- Optimizing these technologies on their design and operation on DHC networks can also yield reduced costs apart from their environmental benefits and provide electricity flexibility services ^[5,6]



Challenge:

- How beneficial is the integration of Heat Pumps in DHC networks ?
- From which factors is their integration influenced and how their potential is further unlocked ?

1) IEA - Climate Watch, "Historical GHG emissions: Greenhouse gas emissions by sector", 1990–2023.
2) European Commission, "Heating and cooling." – Heating and cooling efficiency, 2025.
3) L. Cheng, Z. Yu, S. Xia, S. Li, Y. Li, H. Zhang, B. Li, S. Zhang, Z. Liu, and W. Zheng, "Evaluation and optimization of heat pump combined district heating system: A case study of China," *Energies*, vol. 15, no. 20, 2022.
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5) R. Lund, D. D. Ilic, and L. Trygg, "Socioeconomic potential for introducing large scale heat pumps in district heating in Denmark," vol. 139, pp. 219–229.
6) N. Javanshir, S. Syri, S. Tervo, and A. Rosin, "Operation of district heat network in electricity and balancing markets with the power-to-heat sector coupling," *Energy*, 2023.

Background

A)

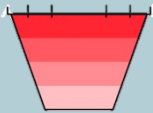
Key industry milestones on the topic



Industry state-of-the-art:



Everllence's mega heat pump project in Esbjerg, Denmark, 70 MW single unit ^[7]



Hot water pit seasonal thermal energy storage (TES) in Dronninglund (60,000 m³) and in Høje Taastrup (70,000 m³), Denmark ^[8,9]



Helsinki's DHCN operated by Helen (Eiranranta project) delivers 90 MW heating and 60 MW cooling through HPs ^[10]



Stockholm's DHCN operated by Exergi delivers 225 MW heating and cooling through HPs ^[11]

7) MAN Energy Solutions. *Mega Heat Pump Delivers First Heat in Esbjerg*.

8) I. Sifnaios, G. Gauthier, D. Trier, and J. Fan, "Dronninglund water pit thermal energy storage dataset" *Solar Energy*, vol. 251, pp. 68–76, Feb. 2023..

9) I. Sifnaios, and A. R. Jensen, "Performance analysis of the Høje Taastrup water pit thermal energy storage," *Journal of Energy Storage*, vol. 131, 2025.

10) Helen, "Eiranranta heat pump plant."

11) Smart City Sweden, "Stockholm innovates district heating with new solutions and renewable sources | news."

B)

Energy System Modeling



Key literature contributions and takeaways :

1. *MILP formulation dominates* ^[12]
2. *LSHP is the most suitable technology for future district networks* ^[13]
3. *Joint design and operation optimization is essential for evaluating multi-energy systems* ^[14]
4. *HP-TES-DHC potential is unlocked by participation in flexibility services* ^[15]

C)

Flexibility services participation



Modeled with :

1. *Stochastic optimization and price clusters* ^[16]
2. *Robust optimization (worst case scenario) and Conditional Value at Risk* ^[17]

12) D. Wang, X. Li, "Advancing the thermal network representation for the optimal design of distributed multi-energy systems," *Frontiers in Energy Research*, vol. 9, p. 668124, 2021.

13) H. Pieper and A. Volkova, "Method of linear approximation of COP for heat pumps and chillers based on thermodynamic modelling and off-design operation," *Energy*, vol. 230, p. 120743, 2021.

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17) Yvet Renkema, "Conformal prediction for stochastic decision-making of PV power in electricity markets," *"Electric Power Systems Research"*, 2024.

**Literature gap:**

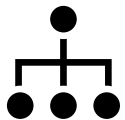
- Regional variability inefficiently explored
- Balancing market participation limited to regional characteristics and market sequential optimization

**Novelty:**

- Cross-regional and climatic comparative analysis between two European countries
- New modeling co-optimization approach on the dispatch optimization of a district network participating in aFRR energy provisions



Main Research Question: To what extent can simultaneously heat-pump coupled district heating and cooling networks be cost-effective and decarbonized?

**Sub-Research Questions topics:**

1. LSHP - DHCN **vs** fossil fuel-based district-only supply strategies
2. Influence of TES presence to the network
3. Market participation in aFRR as additional operational revenue streams



(Zurich)

vs



(Amsterdam)



- Introduction
- **Methodology**
- Results
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Methodology overview – Two stage MILP energy optimization

$$\begin{aligned} & \min_{x,y} (\mathbf{c}^T \mathbf{x} + \mathbf{d}^T \mathbf{y}) \\ \text{s.t.} \\ & A\mathbf{x} + B\mathbf{y} = \mathbf{b} \\ & \mathbf{x} \geq 0 \in \mathbb{R}^{N_x}, \quad \mathbf{y} \in \{0, 1\}^{N_y} \end{aligned}$$

Operational decision variables (MW):

- Input Energy Rate (U)
- Output Energy Rate (V)
- Stored Energy Rate (E)
- Charging/Discharging Energy Rate (U_c, V_d)
- ON/OFF scheduling (y)

Each variable is optimized in every timestep
1st & 2nd stage

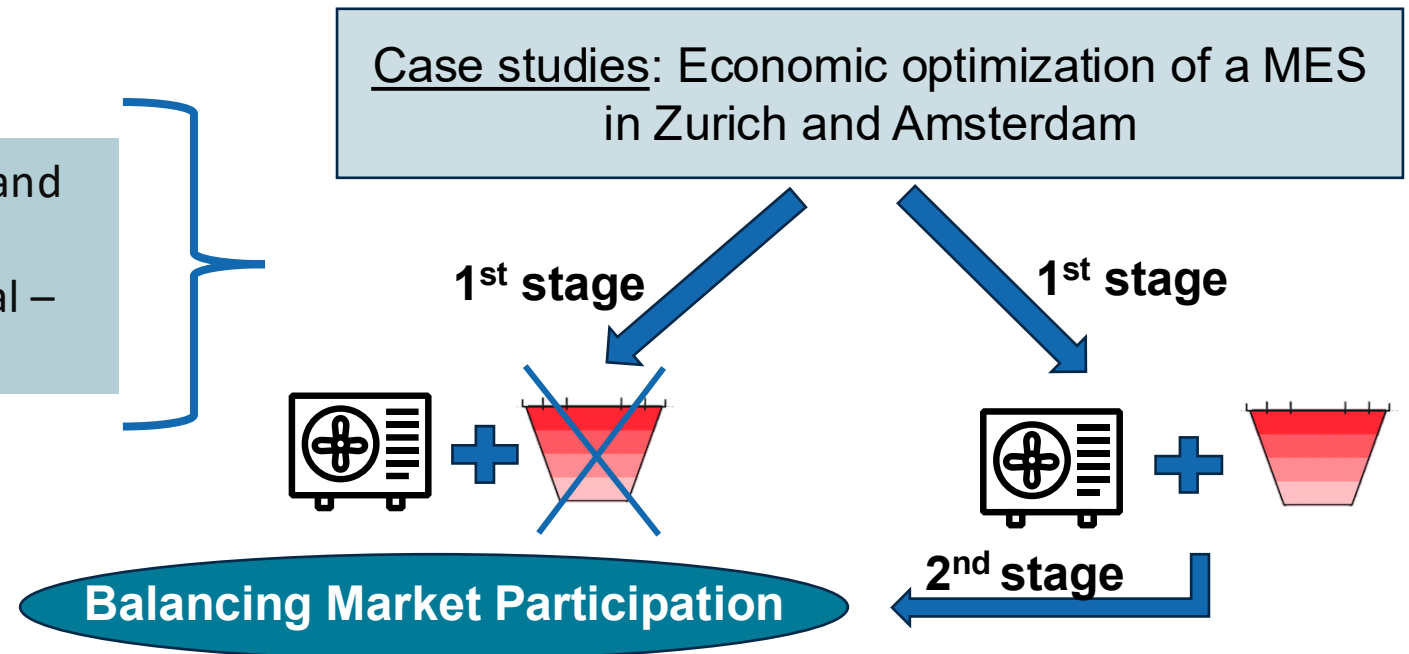
Design decision variables (MW):

- Technology size (P)
 - Technology selection (b)
- Each variable is optimized only once
1st stage

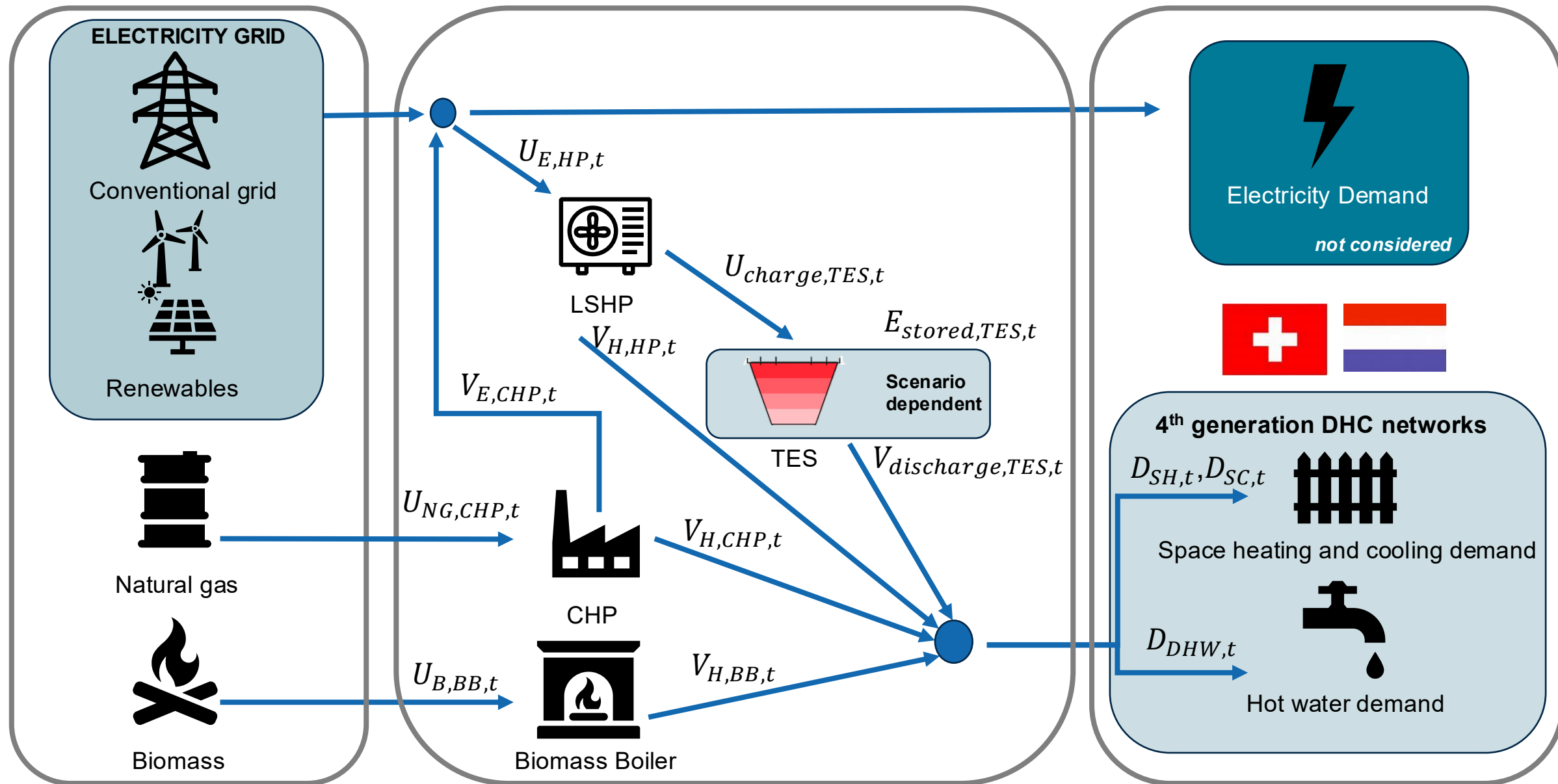
- Continuous variables (X)
- Binary variables (Y)

- ❑ **1st stage: Deterministic optimization** (design and baseline operation – 1h resolution)
- ❑ **2nd stage: Stochastic optimization** (operational – 15min resolution)

Case studies: Economic optimization of a MES in Zurich and Amsterdam



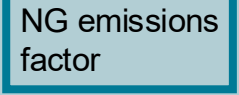
Proposed Multi-Energy System (MES) model



Features of the MILP energy optimization problem

<u>Features of the optimization problem</u>	<u>Chosen attributes for each feature relevant to the thesis work</u>	<u>Remarks</u>
Decision variables (Design and operation)	Selection and size of technologies coupled with their operation for a given set of technologies	MES system
Time dimension	Time-dependent	1h resolution (1 st stage) and 15-min resolution (2 nd stage)
Space Dimension	Single node	One aggregated heating and cooling demand
Uncertainty	Deterministic and stochastic optimization	5 aFRR energy activation price scenarios
Objective function	Single objective	TAC (1 st stage) and TEAC (2 nd stage)

Input data and exogenous parameters (1/2)

	Data sources	Data processing and final inputs
Heating and cooling demand	 + 2025  + Annual heating and cooling consumption	- Generic profiles for SH, SC and DHW - Normalization of profiles (residential and tertiary) - Profile assembly (MWh) $[\forall t \in \mathcal{T}]$
Technology Efficiencies	- Sea/lake water temperature profiles and DHCN supply temperatures) - Empirical regression formula for COP ^[18]	- Heating and Cooling COP $[\forall t \in \mathcal{T}]$ $\eta_k, \eta_{c,TES}, \eta_{d,TES}, \lambda$
LSHP DA electricity procurement	 +  	2025 DA electricity market prices $\left(\frac{\text{€}}{\text{MWh}}\right)$ 2025 Grid Tariff $\left(\frac{\text{€}}{\text{MWh}}\right)$ - DA procurement price = DA price + Grid Tarif $\left(\frac{\text{€}}{\text{MWh}}\right)$ $[\forall t \in \mathcal{T}]$
CHP input fuel prices	 +  * 	2025 TTF natural gas price $\left(\frac{\text{€}}{\text{MWh}}\right)$ 2025 EU ETS carbon allowances $\left(\frac{\text{€}}{\text{ton}_{CO_2}}\right)$ - Natural gas burning emission factor $\left(\frac{\text{ton}_{CO_2}}{\text{MWh}}\right)$ - Input price = TTF + (EU ETS * Emission Factor) $\left(\frac{\text{€}}{\text{MWh}}\right)$ $[\forall t \in \mathcal{T}]$

18) M. Jesper, F. Schlosser, F. Pag, T. G. Walmsley, B. Schmitt, and K. Vajen, "Large-scale heat pumps: Uptake and performance modelling of market-available devices," *Renewable and Sustainable Energy Reviews*, vol. 137, p. 110632, 2021.

Input data and exogenous parameters (2/2)

	Data sources	Data processing and final inputs
Balancing prices	 	• 2016-2025 historic data on upward/downward energy flexibility $\left(\frac{\text{€}}{\text{MWh}}\right)$ $[\forall t \in \mathcal{T}_{15}]$ • K-Medoids clustering
Costs and sizing	• State of the art industrial applications • Public data on construction costs and sizes	• Investment cost $CAPEX_k \left(\frac{\text{€}}{\text{MW}}\right)$ • Fixed operational and maintenance cost $OPEX_k \left(\frac{\text{€}}{\text{MW}}\right)$ • Available market sizing $P_{k,min}, P_{k,max}$ (MW) • Minimum load fraction δ_k (%)
GHG emissions		• Input energy carriers direct and indirect GHG emissions $\left(\frac{\text{ton}_{CO_2}}{\text{MWh}}\right)$ $[\forall j \in J]$ • Technologies embedded emissions $\left(\frac{\text{ton}_{CO_2}}{\text{MW}}\right)$ $[\forall k \in K]$

1st and 2nd stage optimization problem formulations

1st stage

Objective Function:

$$\min \quad \text{TAC} = C_{\text{inv}} + C_{\text{fuel}} - R_{\text{elec}}.$$

Global Demand Constraints:



- District heating balance
- District cooling balance

Technology Specific Constraints:



- Design Limits
- Energy conversion performance
- LSHP mode exclusivity
- Min and max energy output
- TES stored energy boundaries
- TES charging/discharging limits
- TES energy balance

2nd stage

Objective Function:

$$\min \quad \text{TEAC} = C_{\text{fixed}} + \mathbb{E}[C_{\text{ops}}].$$

$$\mathbb{E}[C_{\text{OPS}}] = \sum_{s \in S} \rho_s \cdot (C_{\text{base},s} - R_{\text{base},s} - \mu \cdot R_{\text{bal,up},s} + \nu \cdot C_{\text{bal,down},s})$$

Global Demand Constraints:



- District heating balance
- District cooling balance

Technology Specific Constraints:



- Energy conversion performance
- LSHP mode exclusivity
- Min and max energy output
- BM bidding boundaries
- BM participation direction exclusivity
- TES stored energy boundaries
- TES charging/discharging limits
- TES energy balance

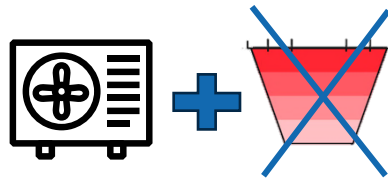
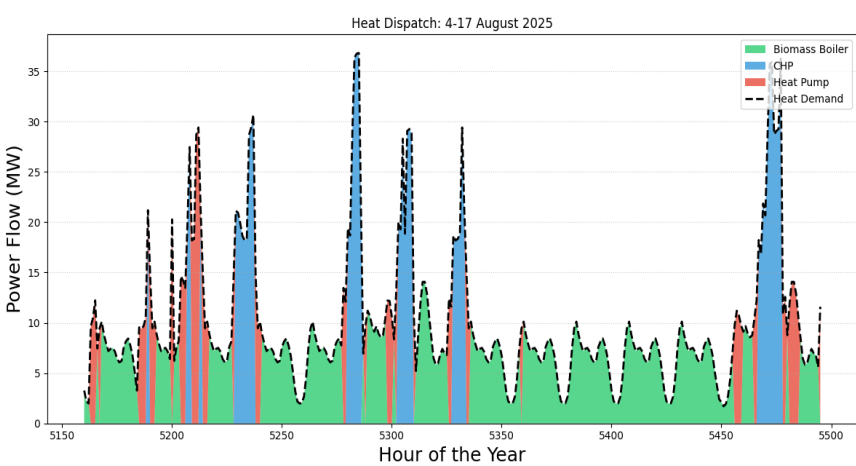
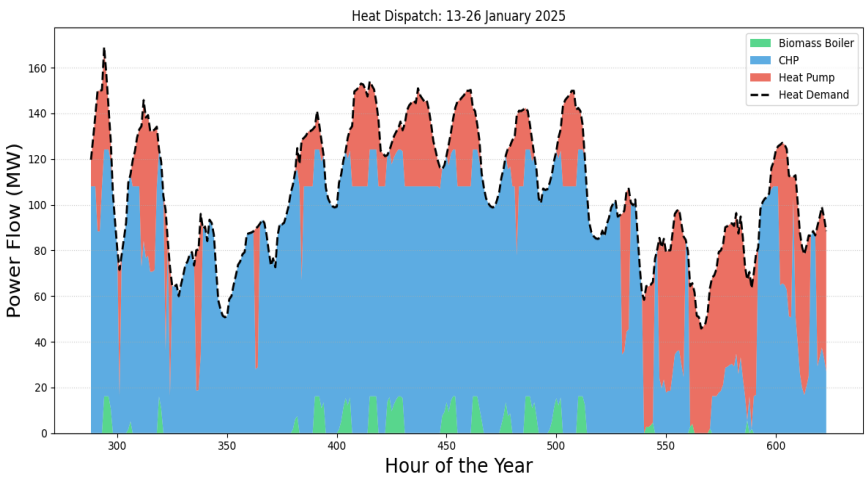


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Brownfield Approach

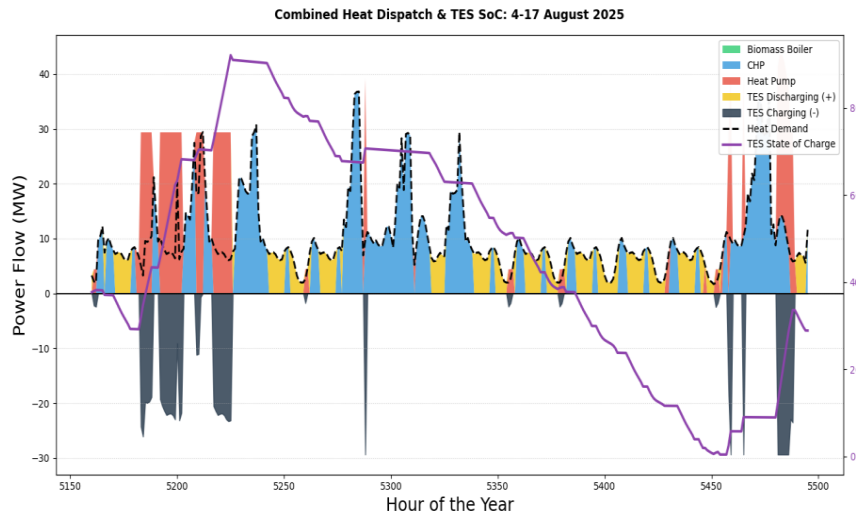
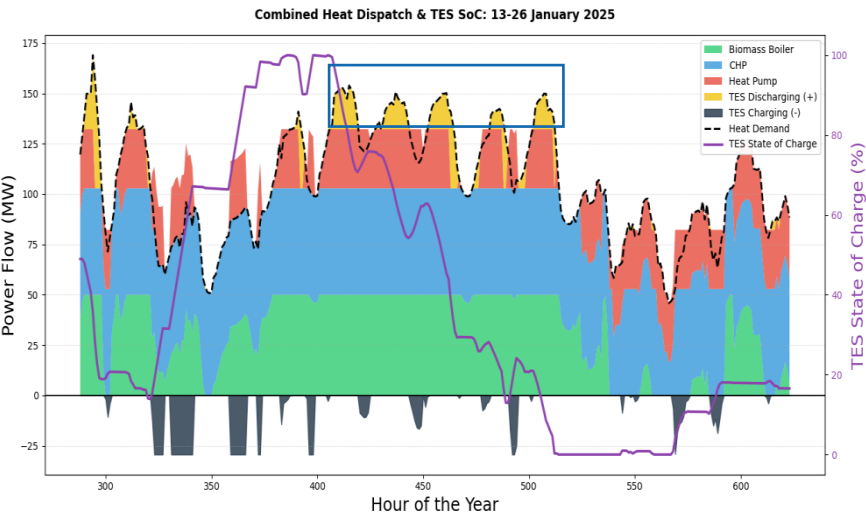
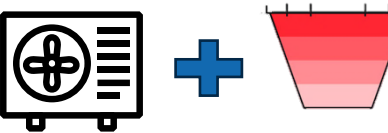
Design and Heat Dispatch operation optimization for 2025 – 1st Stage

Amsterdam



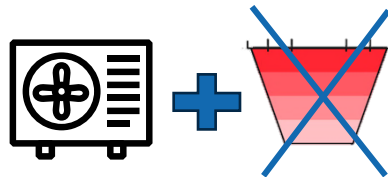
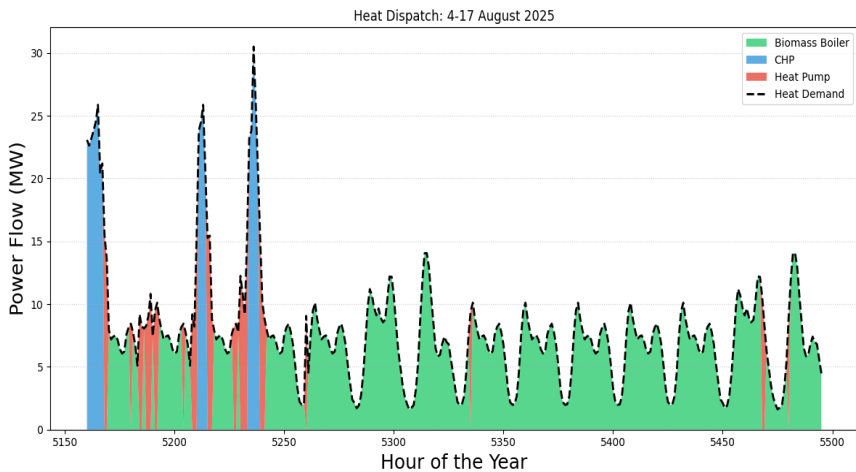
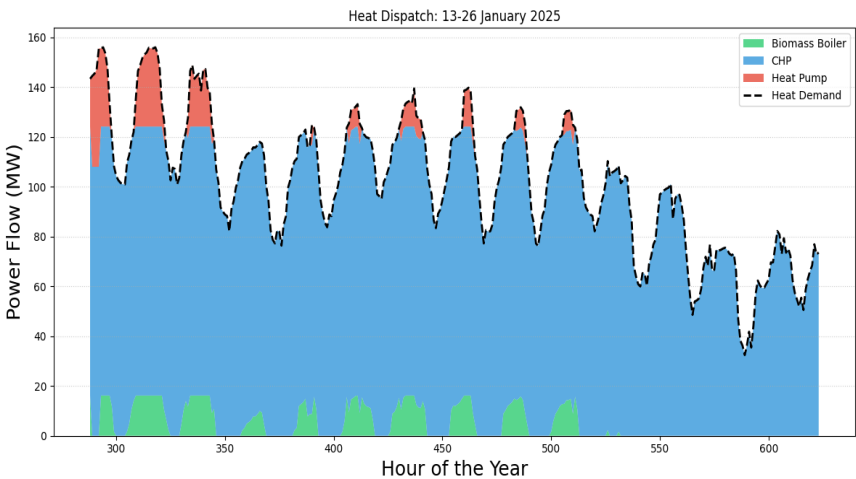
Optimization Results

	BASE	NO TES	WITH TES
BB (MW)	16.1	16.1	50
CHP (MW)	169.8	108	52.9
LSHP (MW)	-	61.5	29.4
TES (m³)	-	-	17440
TAC (M€)	35.6	39.4	29.7
LCODN (€/MWh)	80.81	89.28	67.3
GHG (tonCO ₂)	155.4	135.2	118.2



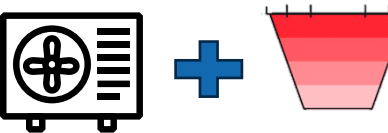
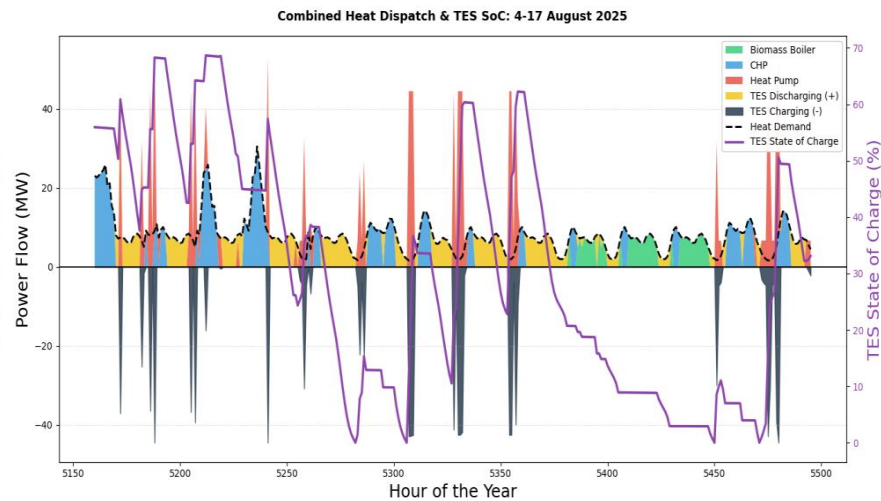
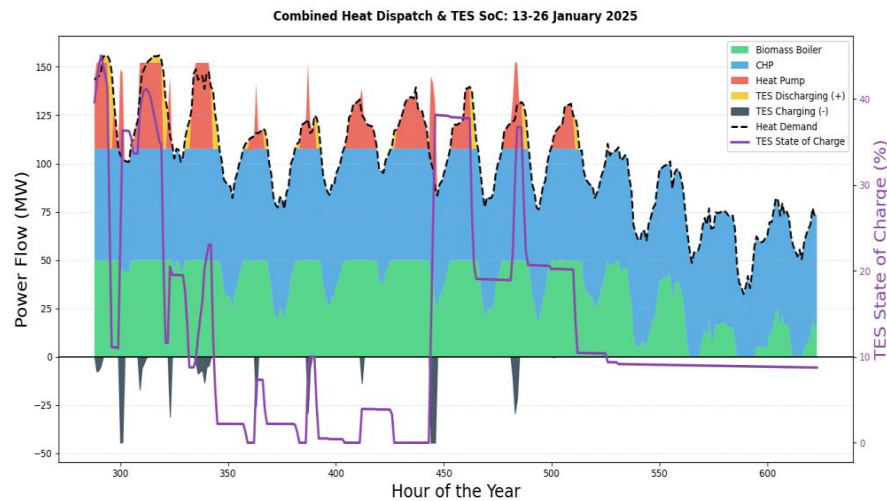
Design and Heat Dispatch operation optimization for 2025 – 1st Stage

Zurich 



Optimization Results

	BASE	NO TES	WITH TES
BB (MW)	16.1	16.1	50
CHP (MW)	161.7	108	57.8
LSHP (MW)	-	53.7	44.5
TES (m³)	-	-	6774
TAC (M€)	38.8	37.9	34.3
LCODN (€/MWh)	86.3	84.35	76.34
GHG (tonCO ₂)	163.8	155.5	120.9



Heat Dispatch operation optimization with BM participation – 2nd stage

Amsterdam

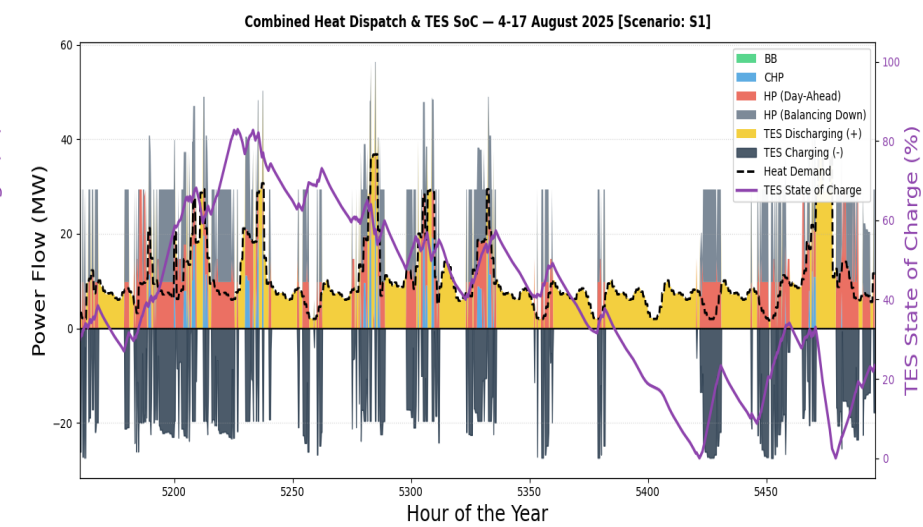
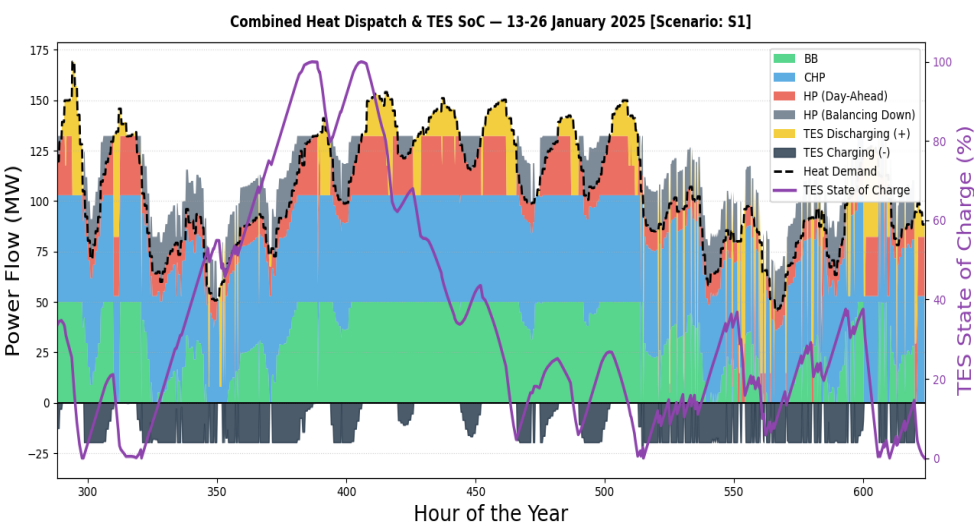
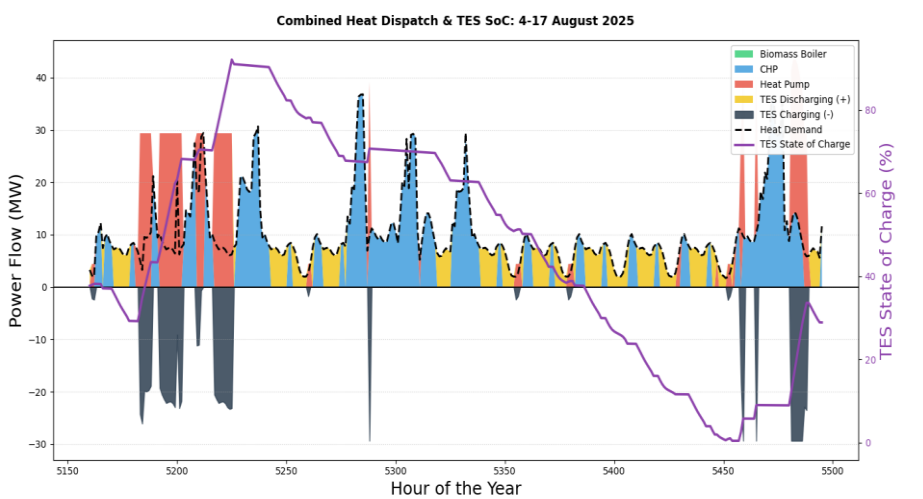
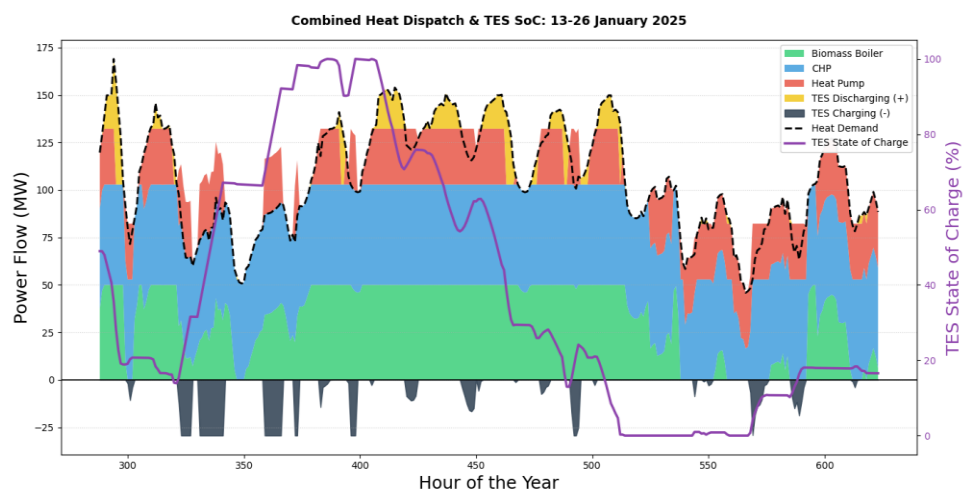
No BM participation

Optimization Results

	No BM,	With BM
BB (MW)	50	50
CHP (MW)	52.9	52.9
LSHP (MW)	29.4	29.4
TES (m ³)	17440	17440
TAC (M €)	29.7	26.1
LCODN (€/MWh)	67.3	59.15
GHG (tonCO ₂)	118.2	108.7

	Offered flexibility
Upward (GWh)	4.2
Downward (GWh)	29.2
Total (GWh)	33.4

Full BM participation



Heat Dispatch operation optimization with BM participation – 2nd stage

Zurich 

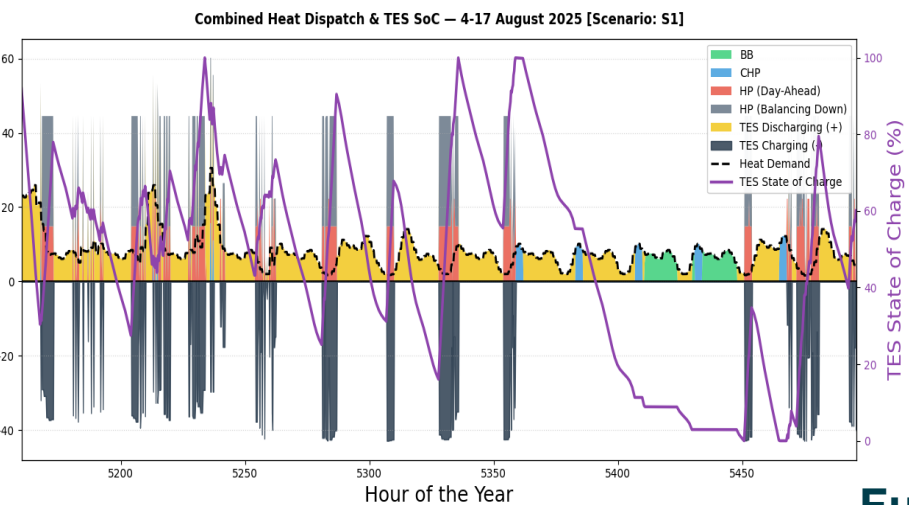
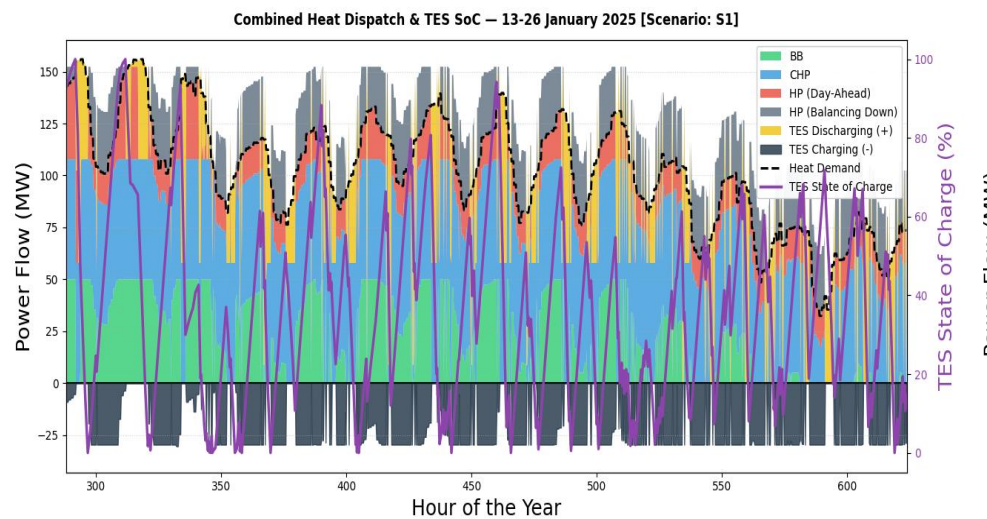
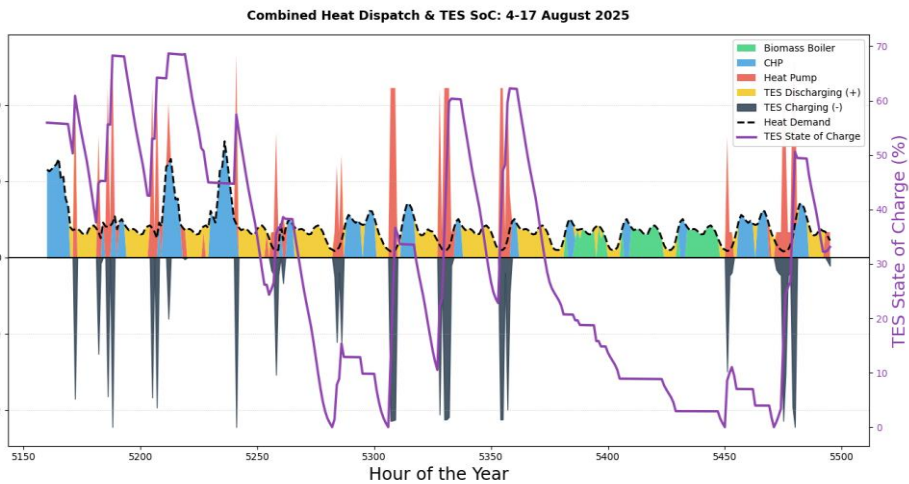
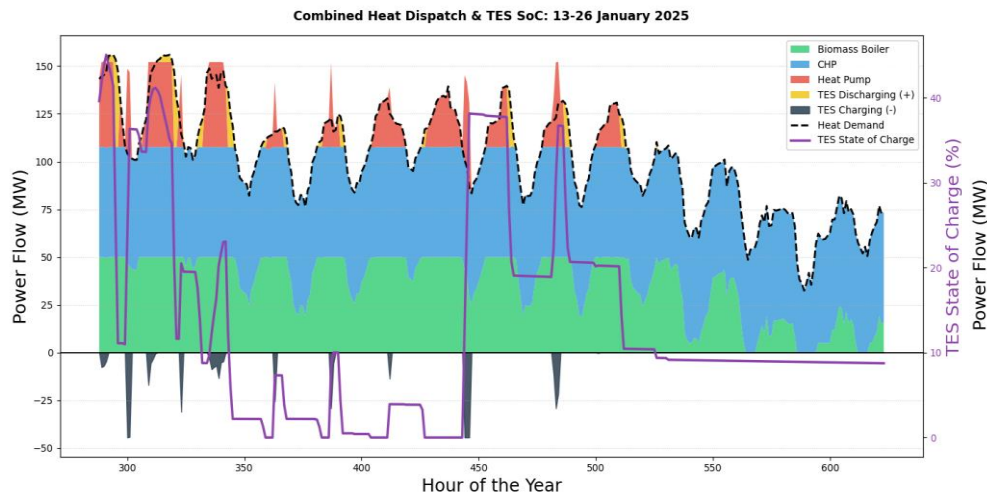
No BM participation

Optimization Results

	No BM,	With BM
BB (MW)	50	50
CHP (MW)	57.8	57.8
LSHP (MW)	44.5	44.5
TES (m ³)	6774	6774
TAC (M€)	34.3	29.9
LCODN (€/MWh)	76.34	66.41
GHG (tonCO ₂)	120.9	89.3

	Offered flexibility
Upward (GWh)	7.4
Downward (GWh)	37.9
Total (GWh)	45.3

Full BM participation



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Conclusions

To what extent can simultaneously heat-pump coupled district heating and cooling networks be cost effective and decarbonized ?

- In both examined networks integration of HPs especially when coupled with TES yields significant economic and environmental benefits under both approaches used

Does participation in BM provide further costs reduction through added operational revenue?

- Full BM participation can lead to expected reduced costs of around 4 M € irrelevant of a brownfield or greenfield approach

How do Zurich and Amsterdam networks compare over the optimization results ?

- Zurich has a higher full BM participation potential thanks to more favorable BM prices
- Zurich has a higher decarbonization potential thanks to “cleaner” electricity grid
- Amsterdam has a lower LCODN because of more favorable DA prices and spreads

Conclusions and Discussion

Limitations of the study

- No complex electricity trading strategies were investigated (interaction between DAM and IDM or other BM segments)
- No BM downward flexibility was used to cover heat demand
- No realistic BM activation bid acceptance ratios were investigated
- No price uncertainty for design optimization and no ramp up/down technology constraints were investigated

Further Work

- Improve the model by inserting uncertainty from 1st stage optimization and provide more complex interaction between different market segments
- Assess the environmental burden of the configuration examined apart only from CO₂ emissions

Thank you

References

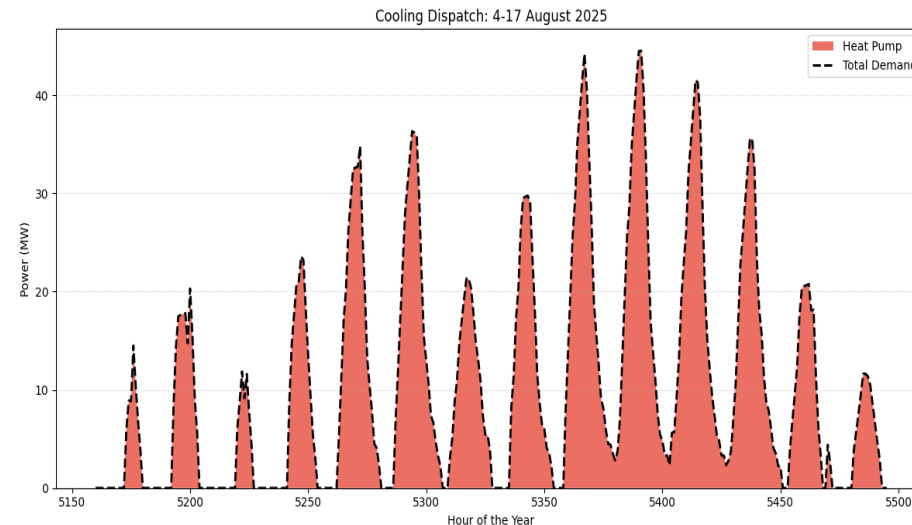
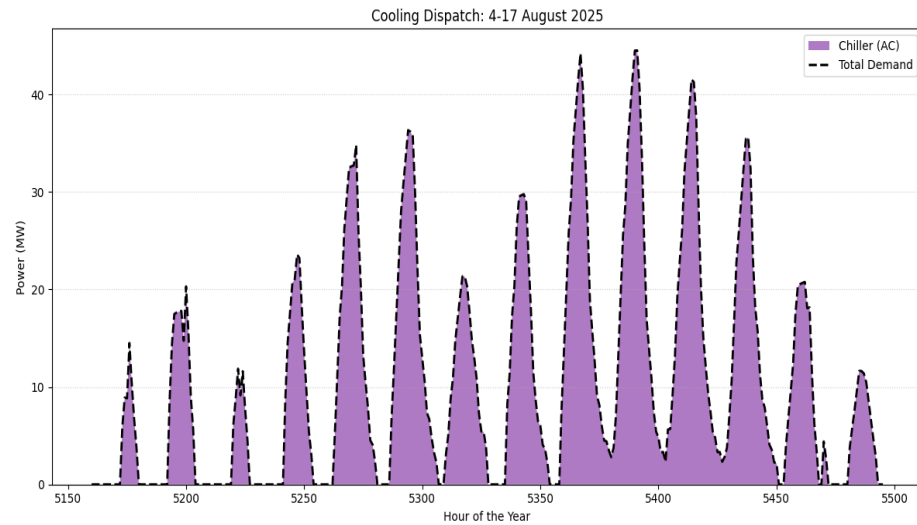
- 1) International Energy Agency - Climate Watch, "Historical GHG emissions: Greenhouse gas emissions by sector", 1990–2023.
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- 17) Yvet Renkema, "Conformal prediction for stochastic decision-making of PV power in electricity markets", "Electric Power Systems Research", 2024.

Appendix

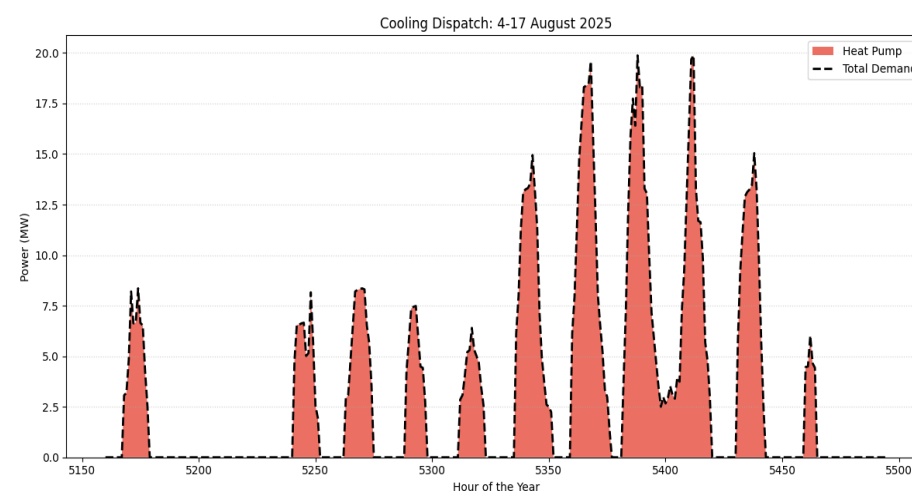
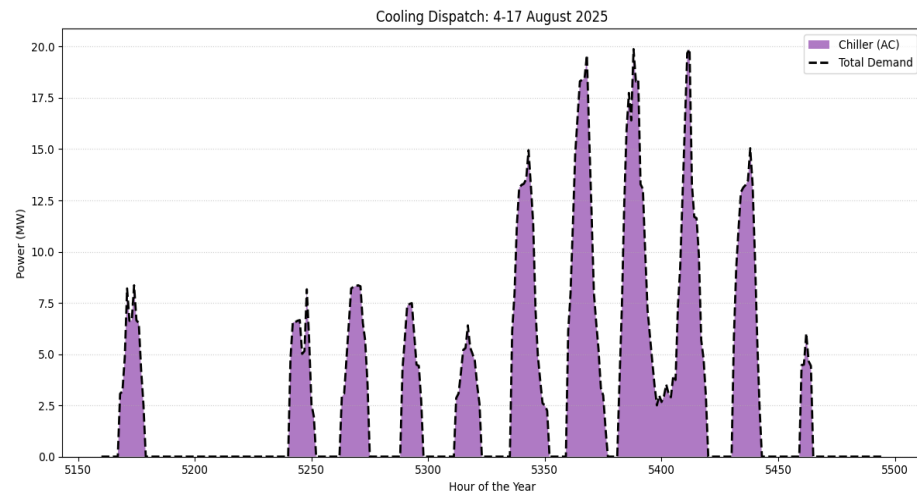
Mathematical optimization: Model sets & Indices

Index / Set	Set Description	Details
$c \in C$	Examined Cities / Networks	Zurich (ZH), Amsterdam (AM)
$j \in J$	Energy Carriers	Natural Gas (NG), Biomass (B), Electricity (E), Heat (H), Cooling (C)
$k \in K$	Energy Generation Technologies	Biomass Boiler (BB), Combined Heat & Power (CHP), Large Scale Heat Pump (HP), Electric Chiller (CH)
$s \in S$	Stochastic Scenarios	Evaluated exclusively within the second-stage optimization
$t \in T / T_{15}$	Temporal Resolution	Hourly intervals ($t \in T$ for 1st stage) 15-minute intervals ($t \in T_{15}$ for 2nd stage)

Cooling dispatch operation optimization



Zurich



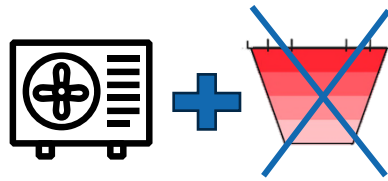
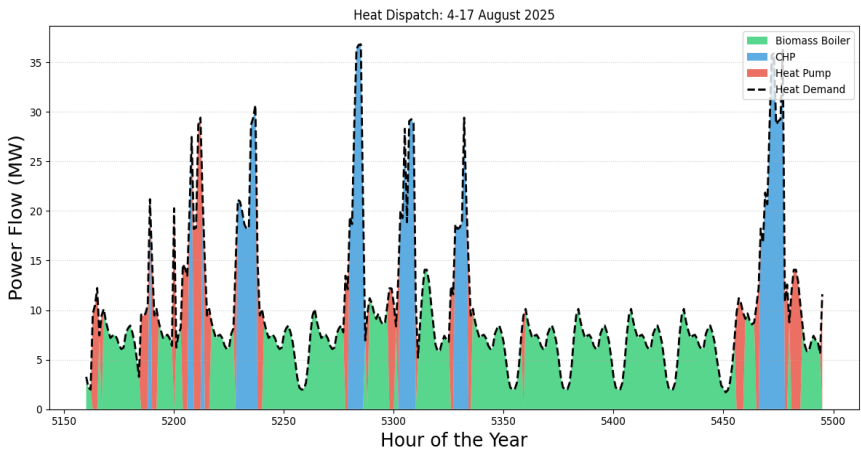
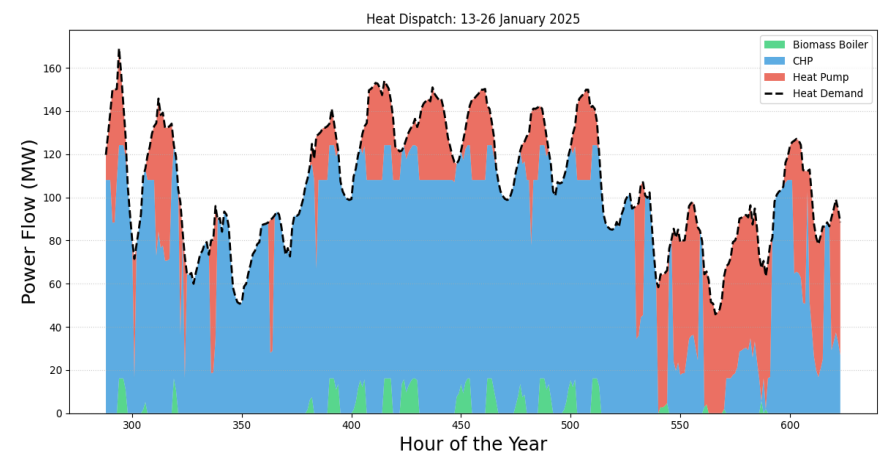
Amsterdam



Greenfield Approach

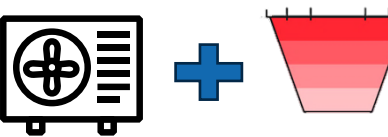
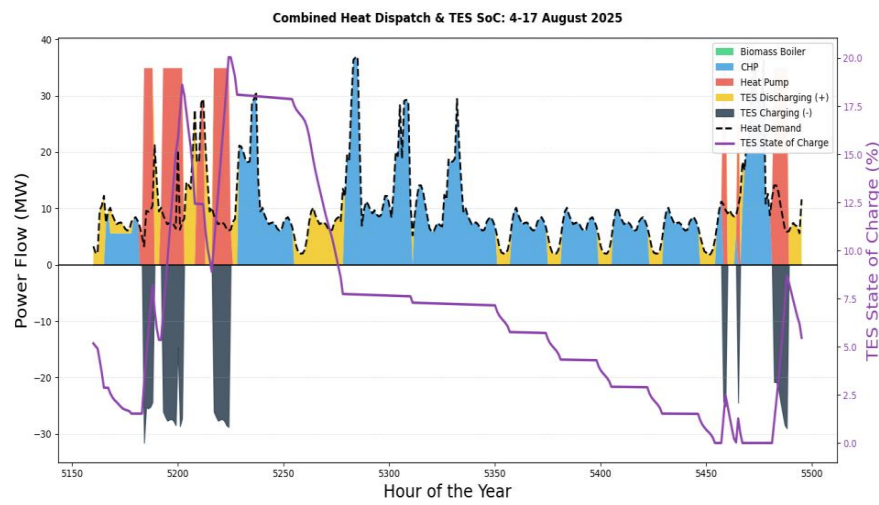
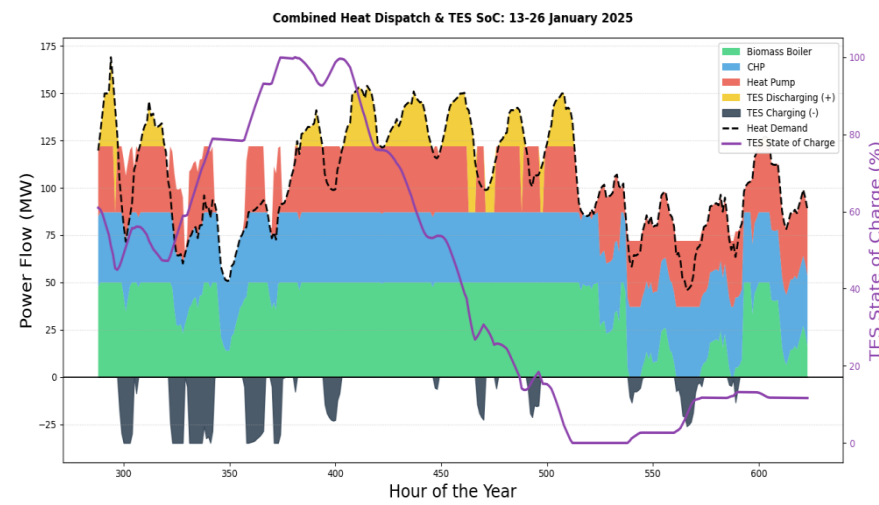
Design and Heat Dispatch operation optimization for 2025 – 1st Stage

Amsterdam



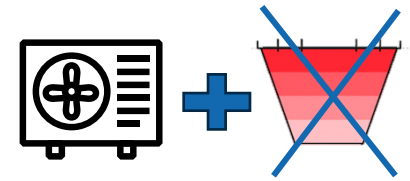
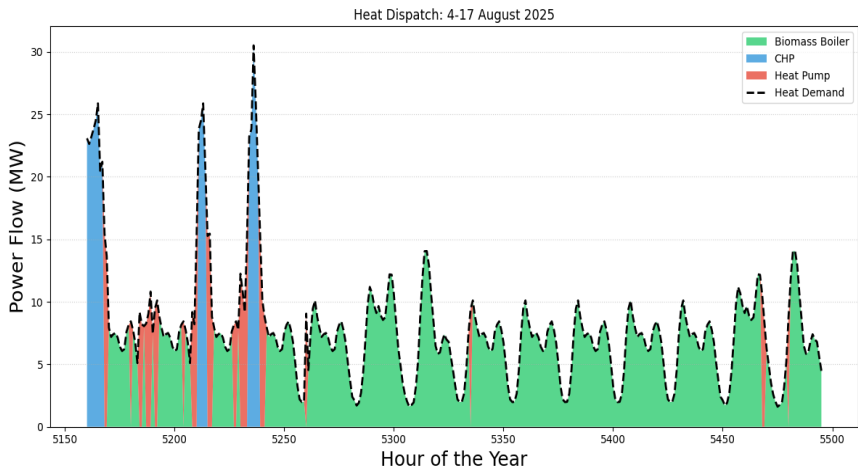
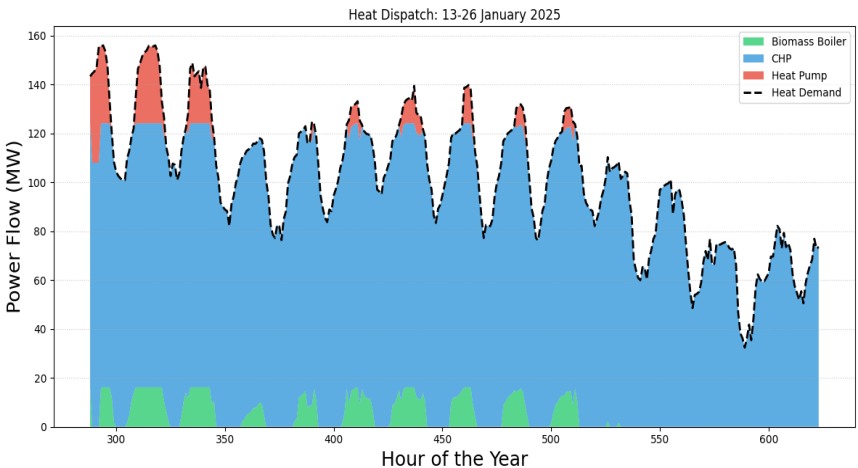
Optimization Results

	BASE	NO TES	WITH TES
BB (MW)	16.1	16.1	50
CHP (MW)	169.8	108	37.1
LSHP (MW)	-	61.5	34.9
TES (m³)	-	-	38142
TAC (M€)	68.2	60.4	41.1
LCODN (€/MWh)	154.7	136.93	93.2
GHG (tonCO ₂)	155.4	135.2	103.4



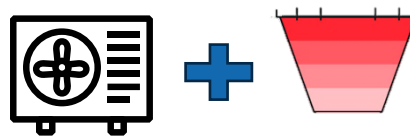
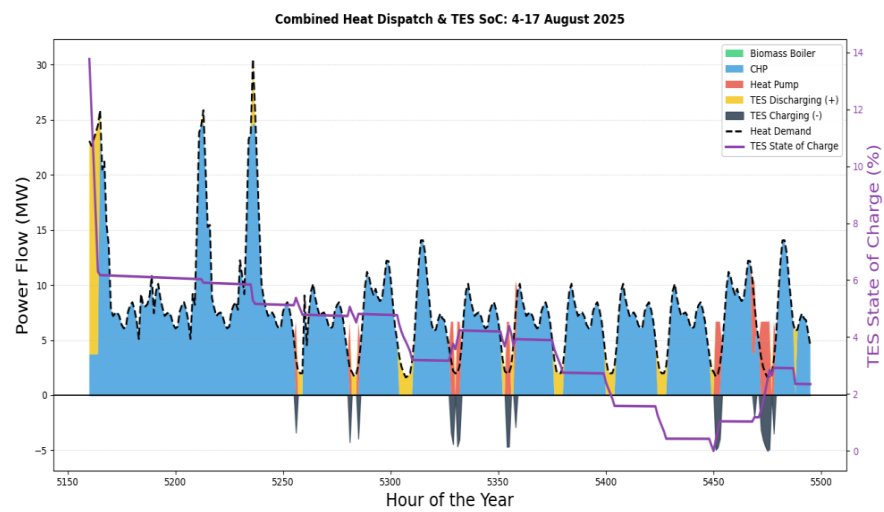
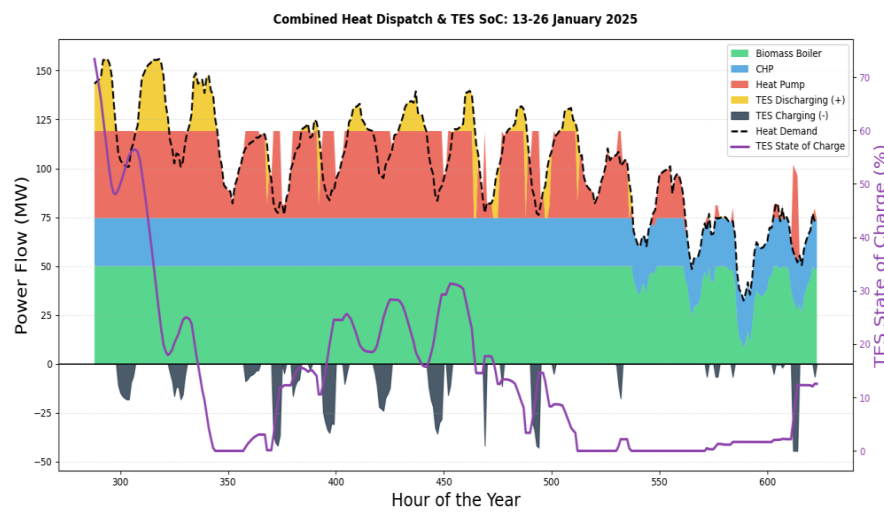
Design and Heat Dispatch operation optimization for 2025 – 1st Stage

Zurich 



Optimization Results

	BASE	NO TES	WITH TES
BB (MW)	16.1	16	50
CHP (MW)	161.7	108	24.6
LSHP (MW)	-	53.9	44.5
TES (m³)	-	-	25489
TAC (M€)	69.9	58.9	42.9
LCODN (€/MWh)	155.39	131.11	95.46
GHG (tonCO ₂)	163.8	155.4	69.5



Heat Dispatch operation optimization with BM participation – 2nd stage

Amsterdam

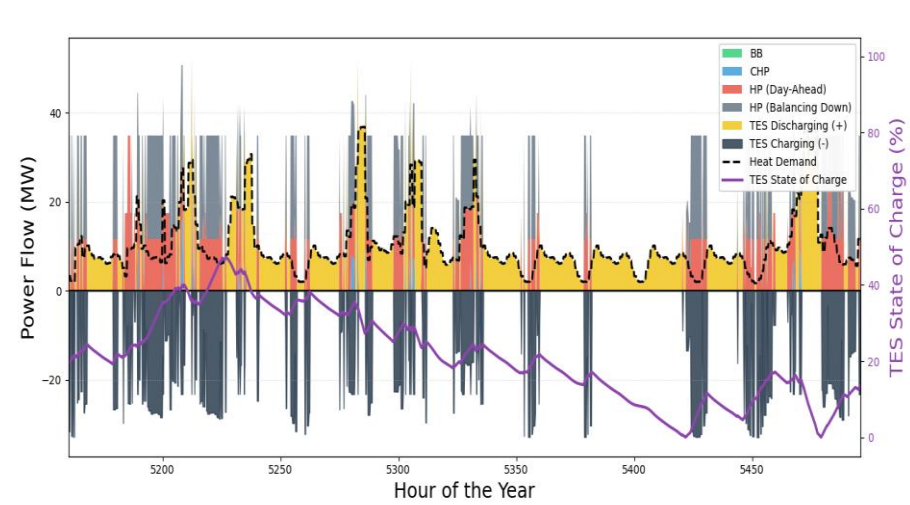
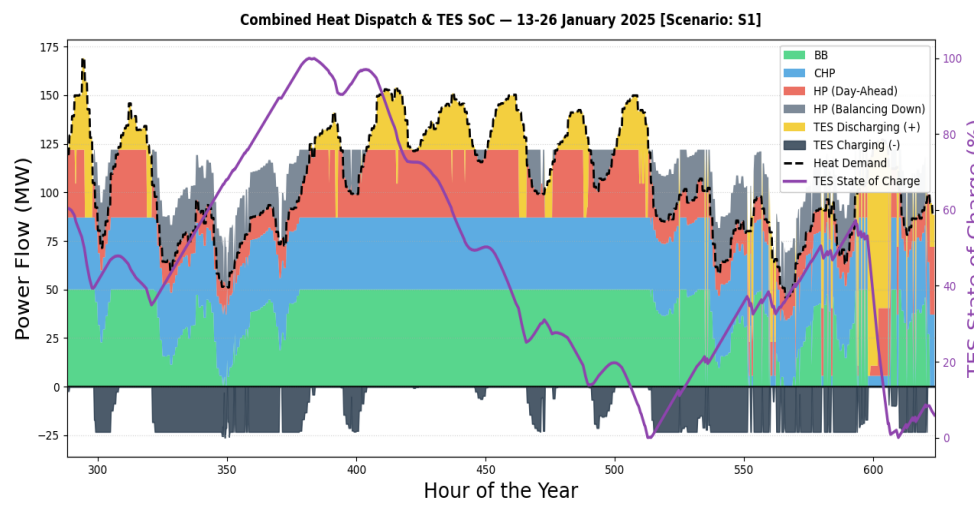
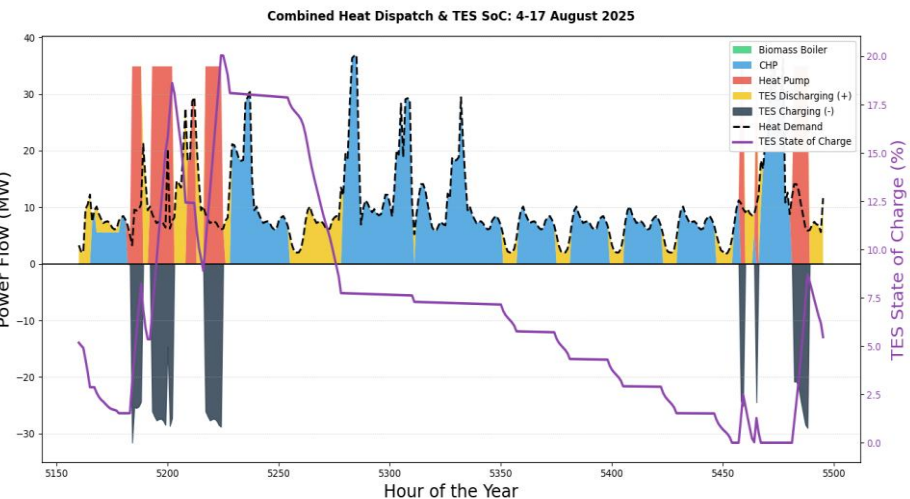
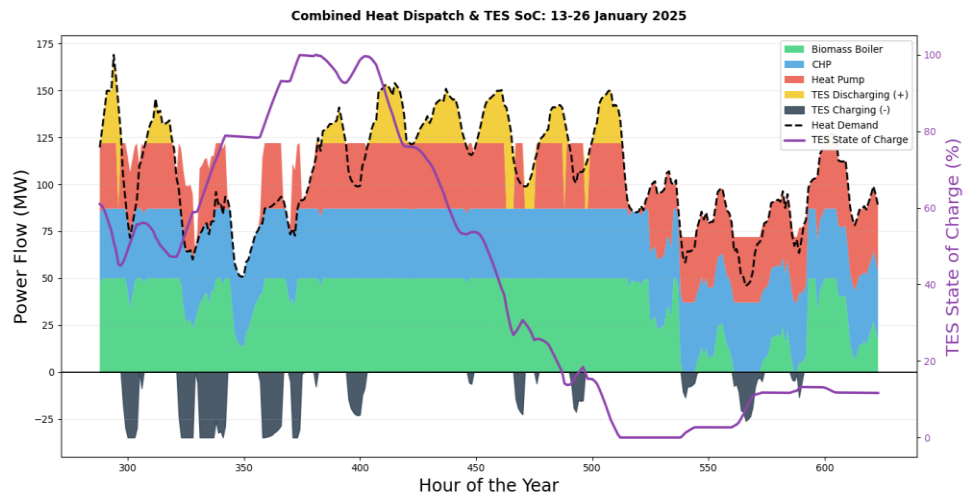
No BM participation

Optimization Results

	No BM,	With BM
BB (MW)	50	50
CHP (MW)	37.1	37.1
LSHP (MW)	34.9	34.9
TES (m³)	38142	38142
TAC (M€)	41.1	37.1
LCODN (€/MWh)	93.2	84.08
GHG (tonCO ₂)	103.4	93.2

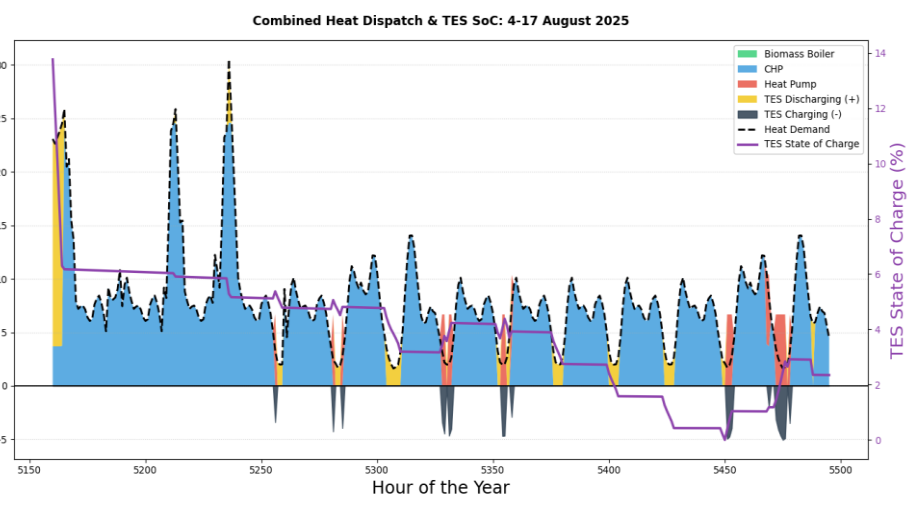
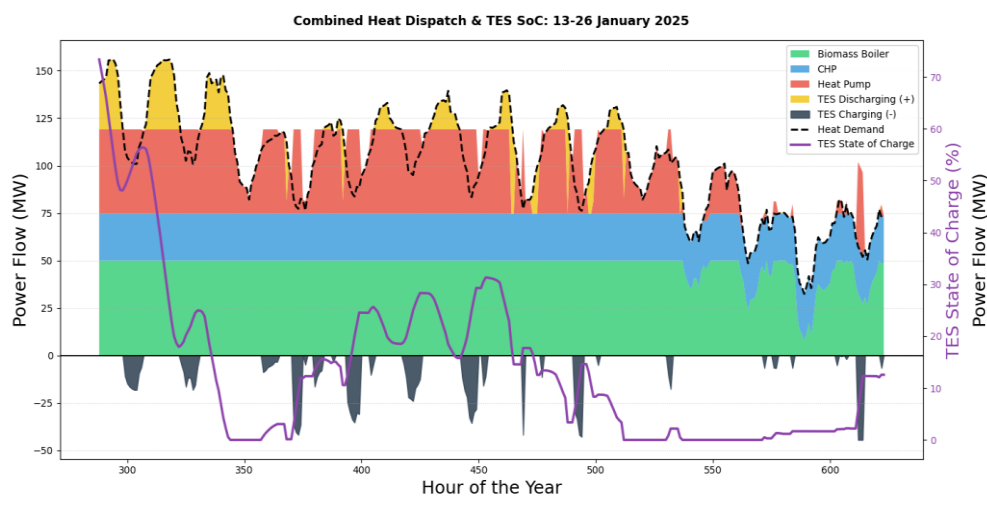
	Offered flexibility
Upward (GWh)	4.8
Downward (GWh)	31.5
Total (GWh)	36.3

Full BM participation



Heat Dispatch operation optimization with BM participation – 2nd stage

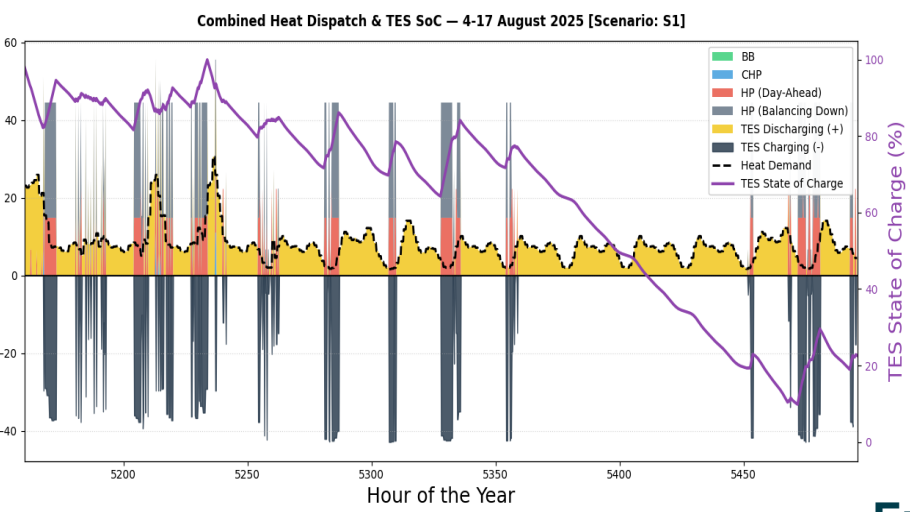
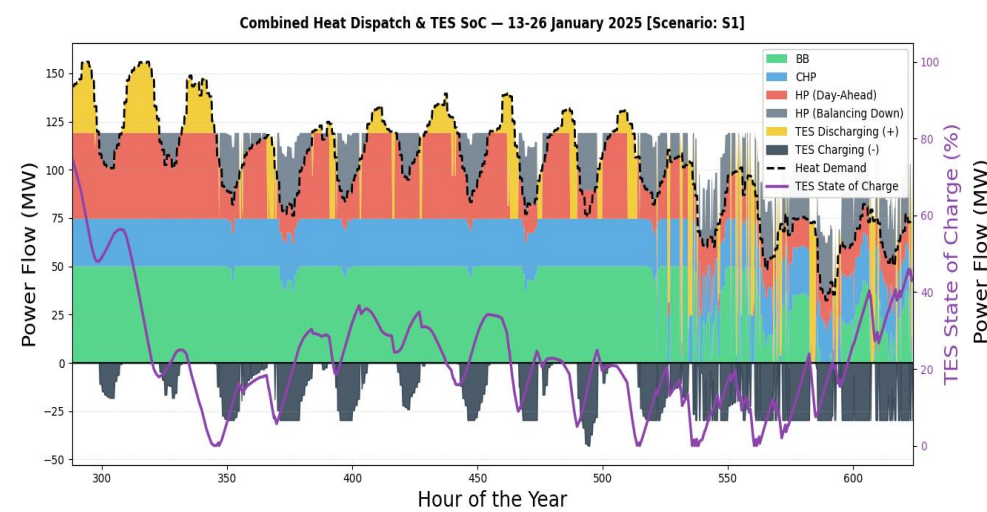
Zurich 



No BM participation

Optimization Results

	No BM,	With BM
BB (MW)	50	50
CHP (MW)	24.6	24.6
LSHP (MW)	44.5	44.5
TES (m ³)	25489	25489
TAC (M €)	42.9	38.2
LCODN (€/MWh)	95.46	84.85
GHG (tonCO ₂)	69.5	51.5

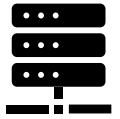


	Offered flexibility
Upward (GWh)	7.5
Downward (GWh)	36.9
Total (GWh)	44.4

Full BM participation

K-Medoids for addressing electricity markets price uncertainty

10 years of
historic price data
(2015-2025)



**K-medoids
clustering**

- N price scenarios ($N \leq 10$)
- Probability = $P = \frac{N_K}{N}$ (historic frequency of occurrence)

$$\min \sum_{i=1}^N \sum_{j=1}^K z_{ij} d(x_i, m_j)$$

Subject to
$$\sum_{j=1}^K z_{ij} = 1 \quad \forall i$$

- x_i : historic electricity price scenario (actual year with 35040 15-min values)
- m_j : medoid of cluster j
- $d()$ is the Euclidean distance function
- $z_{ij} \in \{0,1\}$, 1 if year i belongs to cluster j
0 otherwise

Why K-medoids?

Electricity prices exhibit

- High Volatility
- Extreme spikes
- Non-normal distributions

Clustering based on real data years

Compared to K-means and other clustering techniques ?

- More robust to outliers than K-Means
- It generates representative stochastic scenarios